Module 7: Classsification

Part 2: LDA & QDA

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Recap

When does logistic regression break down?

- When there is substantial separation between the two classes, parameter estimates for the logistic regression model are unstable.
  - In some cases, when there is perfect separation, the estimates cannot converge.

- If the distribution of the predictors X is approximately normal in each of the classes and the sample size is small, then other models may be more accurate than logistic regression.

- When p > n (for example, in image classification).

Bayes classifier

Bayes rule: $$\Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}.$$

- A Bayes classifier is a rule that assigns an observation x to the class with largest pk(x).
- The problem is that it is difficult to estimate f_k.
- Three classifiers are suggested to approximates the Bayes classifier with different estimates of f_k :
  - linear discriminant analysis (LDA)
  - quadratic discriminant analysis (QDA)
  - naive Bayes

Linear Discriminant Analysis (LDA) when p = 1

We use Bayes Theorem to approximate $P(Y = k \mid X_1), k = 1, \ldots, K$.

$$P(Y = k \mid X_1) = \frac{P(X_1 \mid Y = k)P(Y = k)}{P(X_1)}$$

LDA (p = 1)

- We assume $P(X_1 \mid Y = k)$ is a **normal distribution**.
- Normal distribution's pdf: $\frac{1}{\sigma_k\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma_k^2}(x-\mu_k)^2\right)$
  - μ_k is the true mean for the k th class.
  - σ_k^2 is the true variance for the k th class.

- We make an additional assumption that $\sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$. We denote this common variance as σ^2.

LDA (p = 1)

$$P(Y = k \mid X_1) = \frac{P(X_1 \mid Y = k)P(Y = k)}{P(X_1)}$$

Skipping technical details. If you

- Plug normal distribution into $P(X_1 \mid Y_k)$.
- Denote $P(Y = k) = \pi_k$.
- Take log and rearrange terms.

For each class, we compute a score $\delta_k(x)$, and assign the observation to the class with the **highest** score.

$$\delta_k(x_1) = x_1 \frac{\mu_k}{\sigma^2} - \frac{\mu_k}{2\sigma^2} + \log(\pi_k)$$

LDA (p = 1)

$$\delta_k(x_1) = x_1 \frac{\mu_k}{\sigma^2} - \frac{\mu_k}{2\sigma^2} + \log(\pi_k)$$

- δ_k is known as the linear discriminant function.
- It gets its name from the fact that is decision boundaries are **linear** in x_1.
- μ_k and σ^2 are parameters and need to be estimated from the data.

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$$

$$\hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

$$\pi_k = n_k/n$$

- Compute decision boundary to assign observations to each class. The decision boundary is the "cutoff point" that separates the two classes. It's the dividing line between different classes.

LDA Example (p = 1)

$$\delta_k(x_1) = x_1 \frac{\mu_k}{\sigma^2} - \frac{\mu_k}{2\sigma^2} + \log(\pi_k)$$

LDA (p > 1)

- We assume that $X = (X_1, X_2, \dots, X_p)$ is drawn from a multivariate Gaussian (multivariate normal) distribution.
- Each class has its own mean vector μ_k (p -dimensional)
- All classes share a common covariance matrix Σ.
- Specifically, we write this as the observations in the k th class are drawn from $N(\mu_k, \Sigma)$.

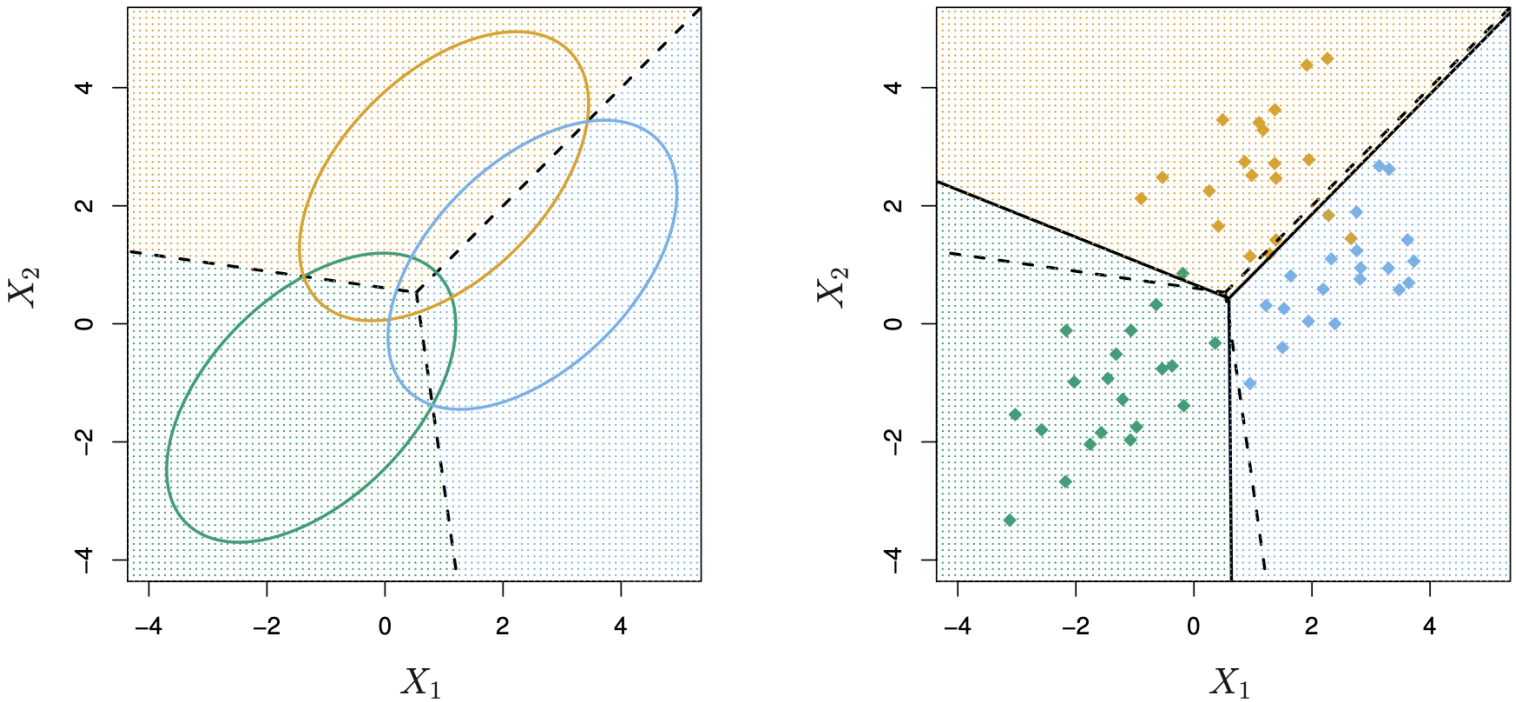
LDA for $p > 1$

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k)$$

- First estimate π_k, μ_k, and Σ.

Compute decision boundary to assign observations to each class.

An example with three classes



LDA: solid black line
Bayes : dashed line

Quadratic Discriminant Analysis (QDA)

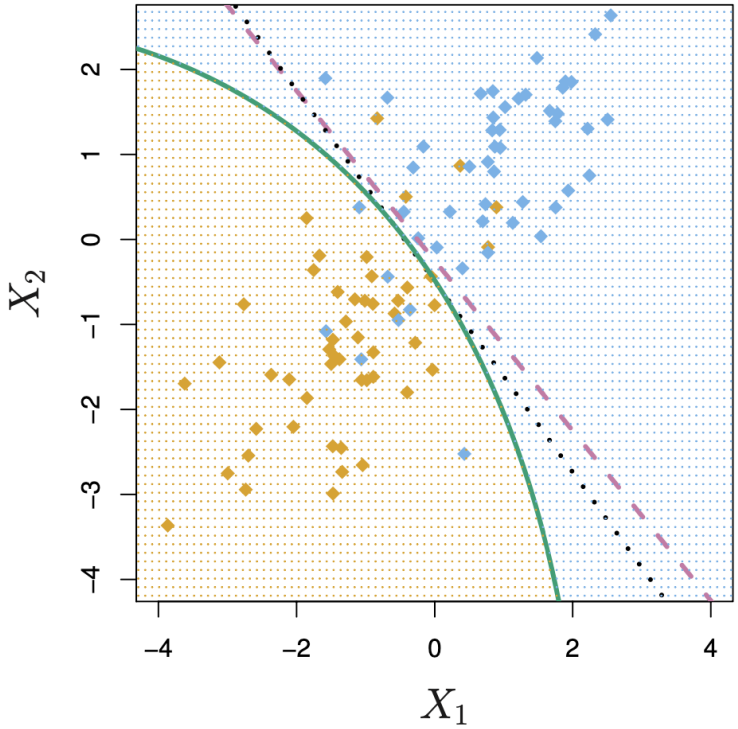
- Common covariance assumption Σ can be quite restrictive.
- QDA relaxes this assumption.
- Each class can have its own covariance matrix:

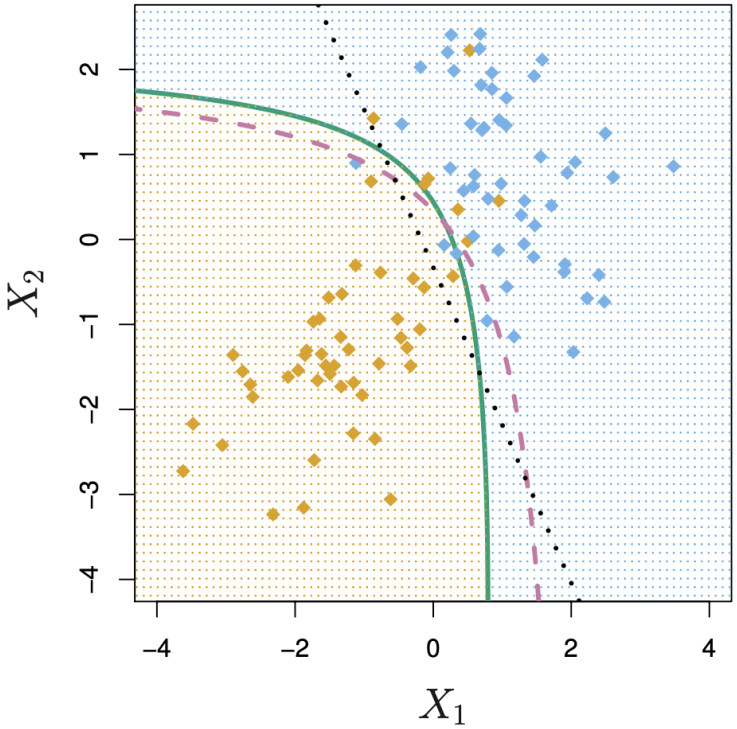
$$X \sim N(\mu_k, \Sigma_k)$$

- Σ_k is the covariance matrix for the k th class.
- Now we need to estimate both μ_k and Σ_k for each class:

$$\delta_k(x) = -\frac{1}{2}x^T\Sigma_k^{-1}x + x^T\Sigma_k^{-1}\mu_k - \frac{1}{2}\mu_k^T\Sigma_k^{-1}\mu_k - \frac{1}{2}\log|\Sigma_k| + \log(\pi_k)$$

Quadratic decision boundaries (QDA)





Why don’t we just always use QDA?

R code

- See script: lda_qda.R